Homework 2 · Part I

Training problem set · Lectures 3 and 4

Deadline: to be announced. Assessed total: 100 points.

Work through this part first, using your notes and showing the requested intermediate steps. Write submitted solutions by hand on paper. Cite external sources and disclose AI assistance; you must be able to explain your work.

This is the ordinary assessed homework. The three assessed homeworks are equally weighted within the course's 20% homework component. Part II is a separate diagnostic mock; do not add or average the two scores. Follow the submission procedure announced by the TA.

For this part, submit Training 1–3 only. The explicitly ungraded further practice has no points or submission requirement. Prices and income are positive, quantities nonnegative and wealth outcomes positive unless stated otherwise.

Training 1. A price tag on a standard of living · 30 points

A consumer has continuous, locally nonsatiated preferences and expenditure function

$$e(p, \bar{u}) = \bar{u}\, p_1^{1/4} p_2^{3/4}, \qquad p \gg 0, \quad \bar{u} > 0.$$

The relevant optima are unique. For a price increase, express CV, EV and the loss of consumer surplus as *positive monetary losses*.

(a) **6 points.** Use Shephard's lemma to recover both Hicksian demands. Verify $p \cdot h = e$.

(b) **8 points.** Invert the expenditure function to obtain $v(p, m)$. Use Roy to recover both Marshallian demands. Check that substituting $\bar{u} = v(p, m)$ into h gives x.

(c) **10 points.** Income is $m = 16$. Prices change from $p^0 = (1, 1)$ to $p^1 = (16, 1)$. Find the initial and final choices and utilities. Calculate compensating variation, equivalent variation and the Marshallian consumer-surplus loss

$$L_{CS} = \int_1^{16} x_1((s, 1), 16)\, ds.$$

Compare the three monetary amounts.

(d) **6 points.** Instead of restoring initial utility, a scholarship administrator makes the initial basket just affordable at the new prices. How much money is added? Compare it with CV and calculate the utility the consumer can now attain after reoptimising.

Training 2. Two shocks, one average · 35 points

Two wealth lotteries have equally likely outcomes:

$$A: \ (36, 100), \qquad B: \ (4, 132).$$

Use Bernoulli indices $b(w) = \sqrt{w}$ and $g(w) = \log w$ on $w > 0$. The certainty equivalent solves $b(CE) = \mathbb{E}[b(W)]$; the risk premium is $\mathbb{E}[W] - CE$.

(a) **6 points.** Calculate both means. Compare their cumulative distribution functions and determine whether either lottery first-order stochastically dominates the other.

(b) **8 points.** Under b, calculate expected utility, the certainty equivalent and the risk premium for each lottery.

(c) **7 points.** Find A's certainty equivalent under g. Compare A with sure wealth 62 under each index. Explain why an increasing change in the Bernoulli index can change lottery rankings.

(d) **6 points.** Calculate absolute and relative risk aversion for b and g. Which index represents greater absolute risk aversion at each wealth?

(e) **8 points.** Use Jensen's inequality to prove that a consumer with an increasing concave Bernoulli index has a nonnegative risk premium. When does strict concavity imply a strictly positive premium?

Further practice, ungraded. Couple the lotteries as follows:

$$\Pr(B = 4 \mid A = 36) = 3/4, \qquad \Pr(B = 4 \mid A = 100) = 1/4.$$

Check B's unconditional distribution and show $\mathbb{E}[B \mid A] = A$. Apply conditional Jensen to show that every increasing concave Bernoulli index weakly prefers A. This establishes second-order stochastic dominance.

Training 3. The umbrella with a loading · 35 points

A household has wealth 100 and faces a loss of 84 with probability $1/4$. Its Bernoulli utility is $b(w) = \log w$. Coverage $I \in [0, 84]$ pays I only if the loss occurs. The premium is cI, paid in either state.

(a) **6 points.** Write both final wealth levels and expected utility. Explain existence and uniqueness of optimal coverage for $c = 1/4$ and $c = 1/2$.

(b) **7 points.** At the actuarially fair price $c = 1/4$, find optimal coverage. Give a Jensen argument, not only a stationary-point calculation.

(c) **10 points.** At $c = 1/2$, find optimal coverage and both final wealth levels. Check that your candidate lies within the allowed interval and is globally optimal.

(d) **12 points.** Keep the loss 84, probability $1/4$ and $c = 1/2$, but replace initial wealth by $W > 84$. Derive optimal coverage as a function of W, including the no-insurance region. Relate the result to absolute risk aversion for log utility.

Homework 2 · Part II

Formative mock examination · Lectures 3 and 4

Deadline: to be announced.

Time: 120 minutes. Diagnostic total: 100 points.

After completing Part I, put it aside and attempt this part in one timed sitting. Write by hand on paper. For an honest diagnostic, use only this question paper, without notes, internet or AI; this is a practice recommendation, not an announcement of the actual final exam's permitted materials.

The TA will mark this part seriously, but mistakes here do **not** lower your final course grade. Keep the original attempt and label corrections made after the timed sitting. Its score is separate from the assessed Part I score.

Ten true/false claims, 5 points each	50
Two goods, one welfare bill	20
Keeping the emergency fund alive	20
Two short answers, 5 points each	10
Total	100

True/false instructions. State a verdict and justify it. For a true claim give a valid argument; for a false one give a fully specified counterexample or a rigorous contradiction. An unsupported label earns no credit. Derivatives and optimisers must be evaluated in the relevant parameter regime.

Welfare convention. For a price increase at fixed income m, report CV, EV and the Marshallian consumer-surplus change as positive *losses*. Distinguish certain wealth from expected utility of random wealth.

Mock 1. True or false · 50 points

Each claim is worth 5 points. For claims 1–5, utility is continuous on $\mathbb{R}^2_+$, preferences are locally nonsatiated, $p \gg 0$, $m > 0$, and targets are attainable and above $u(0)$. Do not assume convexity or differentiability unless stated. At a regular point,

$$S_{ij} = \frac{\partial x_i}{\partial p_j} + x_j \frac{\partial x_i}{\partial m}.$$

1. Concavity without convex tastes

For a fixed target, expenditure is concave in prices even if preferences are not convex.

2. Smooth in income is enough

If $v_m(p, m)$ exists and is strictly positive, the utility-maximising bundle at (p, m) must be unique.

3. Ordinary symmetry

Whenever ordinary demand is single-valued and continuously differentiable,

$$\frac{\partial x_1}{\partial p_2} = \frac{\partial x_2}{\partial p_1}.$$

4. Strict tastes, strict curvature

At a regular point with strictly convex preferences, the Slutsky matrix satisfies $z^\top S z < 0$ for every nonzero z.

5. One gradient, one basket

If $e(\cdot, \bar{u})$ is differentiable at positive prices p, the expenditure problem for that target has a unique minimiser.

For claims 6–10, wealth lotteries have finite positive outcomes. Bernoulli indices are strictly increasing; use risk-aversion coefficients where defined.

6. A change of utility units

Replacing $b(w)$ by $3b(w)+7$ preserves expected-utility preferences, certainty equivalents and Arrow–Pratt absolute risk aversion where defined.

7. Increasing is enough

If $b(w) > 0$, replacing it by $[b(w)]^2$ preserves the ranking of all wealth lotteries, because squaring is increasing on positive numbers.

8. Dominance without concavity

If lottery A first-order stochastically dominates lottery B, every increasing Bernoulli index weakly prefers A; concavity is unnecessary.

9. Decreasing risk aversion prevents full insurance

A consumer with decreasing absolute risk aversion cannot optimally choose full coverage at an actuarially fair insurance premium.

10. One zero premium reveals everything

For a strictly increasing concave Bernoulli index, a zero risk premium for one nondegenerate lottery implies risk neutrality over all positive-wealth lotteries.

Mock 2. Two goods, one welfare bill · 20 points

For $x \in \mathbb{R}^2_+$, utility is

$$u(x) = \frac{x_1 x_2}{x_1 + x_2} \quad \text{when } x_1 + x_2 > 0, \qquad u(0,0) = 0.$$

You may use the indirect utility function without proving it:

$$v(p, m) = \frac{m}{(\sqrt{p_1} + \sqrt{p_2})^2}, \qquad p \gg 0,\ m > 0.$$

(a) **5 points.** Use Roy's identity to recover both demands and verify budget exhaustion.

(b) **5 points.** Use duality to find $e(p, \bar{u})$ for $\bar{u} > 0$. Recover both Hicksian demands using Shephard.

(c) **6 points.** Income is 8 and prices change from $(1, 1)$ to $(4, 1)$. Calculate initial and final utility, CV and EV. Report the two welfare measures as positive monetary losses.

(d) **4 points.** Calculate the Marshallian consumer-surplus loss

$$L_{CS} = \int_1^4 x_1((s, 1), 8)\, ds.$$

Compare it with CV and EV. A useful substitution is $t = \sqrt{s}$.

Mock 3. Keeping the emergency fund alive · 20 points

A freelancer has wealth 144 and faces a loss of 128 with probability $1/2$. Bernoulli utility is $b(w) = \sqrt{w}$. Partial insurance pays coverage I only if the loss occurs; its premium $2I/3$ is paid in either state, and $0 \leq I \leq 128$.

(a) **4 points.** Without insurance, find expected wealth, expected utility, the certainty equivalent and risk premium.

(b) **4 points.** What is the largest *lump-sum premium* the freelancer would pay for full insurance? Compare it with the expected loss.

(c) **4 points.** Write final wealth in both states and the expected-utility objective for partial coverage. Establish strict concavity.

(d) **6 points.** Find optimal coverage, check the interval bounds and calculate wealth in both states at the optimum.

(e) **2 points.** Calculate the certainty equivalent at optimal coverage. Does accepting a lower expected final wealth here contradict expected-utility maximisation?

Mock 4. Two short answers · 10 points

Answer each question in **no more than three sentences**. Each is worth 5 points. These are hypothetical cases; no outside news knowledge is needed.

1. The inflation-proof stipend

After a price change, a scholarship administrator adjusts a student's stipend so that the student's old optimal basket is exactly affordable. Preferences and the available goods stay unchanged.

Does this adjustment preserve *exactly* the old utility? Explain how it differs from Hicksian compensation.

2. The warranty and the raffle

A person buys a warranty priced above the expected repair cost and also a raffle ticket costing 1, paying 10 with probability 0.05 and zero otherwise. The ticket is independent of existing risks and has no entertainment or other nonmonetary benefit; final wealth stays positive.

Can one strictly increasing, globally concave expected-utility index of wealth explain both purchases? Identify an assumption that could be changed to explain the behaviour.